# Chapter 1 VECTORS and SCALARS

**A VECTOR** is a quantity having both magnitude and direction, such as displacement, velocity, force, and acceleration.

Graphically a vector is represented by an arrow $OP$ (Fig.1) defining the direction, the magnitude of the vector being indicated by the length of the arrow. The tail end $O$ of the arrow is called the *origin* or *initial point* of the vector, and the head $P$ is called the *terminal point* or *terminus*.

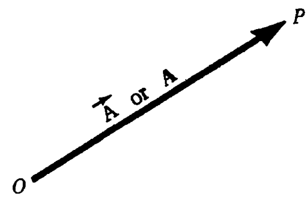



Fig.1

Analytically a vector is represented by a letter with an arrow over it, as $\vec{A}$ in Fig.1, and its magnitude is denoted by $|\vec{A}|$ or A. In printed works, bold faced type, such as **A**, is used to indicate the vector $\vec{A}$ while $|\mathbf{A}|$ or A indicates its magnitude. We shall use this bold faced notation in this book. The vector OP is also indicated as $\overrightarrow{OP}$ or **OP**; in such case we shall denote its magnitude by $\overline{OP}$, $|\overrightarrow{OP}|$, or $|\mathbf{OP}|$.

**A SCALAR** is a quantity having magnitude but no direction, e.g. mass, length, time, temperature, and any real number. Scalars are indicated by letters in ordinary type as in elementary algebra. Operations with scalars follow the same rules as in elementary algebra.

**VECTOR ALGEBRA**. The operations of addition, subtraction and multiplication familiar in the algebra of numbers or scalars are, with suitable definition, capable of extension to an algebra of vectors. The following definitions are fundamental.

1. Two vectors **A** and **B** are *equal* if they have the same magnitude and direction regardless of the position of their initial points. Thus **A** = **B** in Fig.2.
2. A vector having direction opposite to that of vector **A** but having the same magnitude is denoted by −**A** (Fig.3).

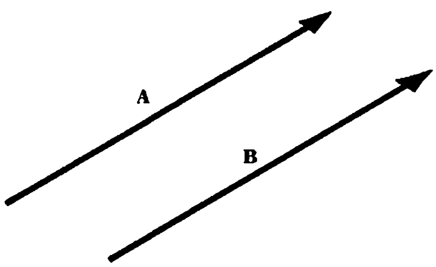



Fig. 2

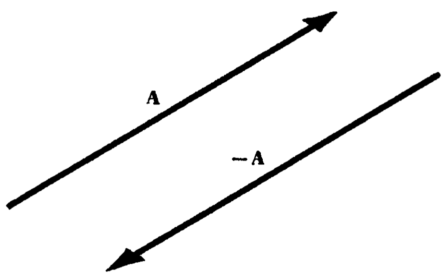



Fig. 3

3. The *sum* or *resultant* of vectors **A** and **B** is a vector **C** formed by placing the initial point of **B** on the terminal point of **A** and then joining the initial point of **A** to the terminal point of **B** (Fig.4). This sum is written **A**+**B**, i.e. **C** = **A**+**B**.

   The definition here is equivalent to the *parallelogram law* for vector addition (see Prob.3).

   Extensions to sums of more than two vectors are immediate (see Problem 4).

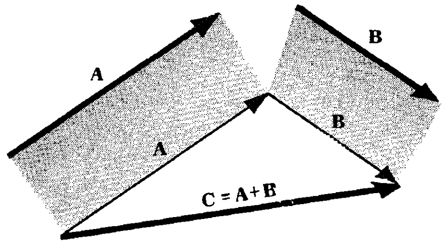



Fig. 4

4. The *difference* of vectors **A** and **B**, represented by **A**−**B**, is that vector **C** which added to **B** yields vector **A**. Equivalently, **A**−**B** can be defined as the sum **A**+(−**B**).

   If **A** = **B**, then **A**−**B** is defined as the *null* or *zero vector* and is represented by the symbol **0** or simply 0. It has zero magnitude and no specific direction. A vector which is not null is a *proper vector*. All vectors will be assumed proper unless otherwise stated.

5. The *product* of a vector **A** by a scalar $m$ is a vector $m$**A** with magnitude $|m|$ times the magnitude of **A** and with direction the same as or opposite to that of **A**, according as $m$ is positive or negative. If $m = 0$, $m$**A** is the null vector.

**LAWS OF VECTOR ALGEBRA.** If **A**, **B** and **C** are vectors and $m$ and $n$ are scalars, then

| | |
|---|---|
| 1. $\mathbf{A}+\mathbf{B} = \mathbf{B}+\mathbf{A}$ | Commutative Law for Addition |
| 2. $\mathbf{A}+(\mathbf{B}+\mathbf{C}) = (\mathbf{A}+\mathbf{B})+\mathbf{C}$ | Associative Law for Addition |
| 3. $m\mathbf{A} = \mathbf{A}m$ | Commutative Law for Multiplication |
| 4. $m(n\mathbf{A}) = (mn)\mathbf{A}$ | Associative Law for Multiplication |
| 5. $(m+n)\mathbf{A} = m\mathbf{A}+n\mathbf{A}$ | Distributive Law |
| 6. $m(\mathbf{A}+\mathbf{B}) = m\mathbf{A}+m\mathbf{B}$ | Distributive Law |

Note that in these laws only multiplication of a vector by one or more scalars is used. In Chapter 2, products of vectors are defined.

These laws enable us to treat vector equations in the same way as ordinary algebraic equations. For example, if **A**+**B** = **C** then by transposing **A** = **C**−**B**.

**A UNIT VECTOR** is a vector having unit magnitude, If **A** is a vector with magnitude $A \neq 0$, then **A**/A is a unit vector having the same direction as **A**.

Any vector **A** can be represented by a unit vector **a** in the direction of **A** multiplied by the magnitude of **A**. In symbols, **A** = A**a**.

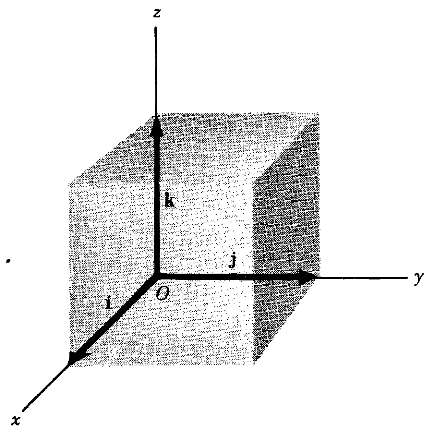



Fig. 5

**THE RECTANGULAR UNIT VECTORS i, j, k.** An important set of unit vectors are those having the directions of the positive $x$, $y$, and $z$ axes of a three dimensional rectangular coordinate system, and are denoted respectively by **i**, **j**, and **k** (Fig.5).

We shall use *right-handed rectangular coordinate systems* unless otherwise stated. Such a system derives its name from the fact that a right threaded screw rotated through 90° from $Ox$ to $Oy$ will advance in the positive $z$ direction, as in Fig.5 above.

In general, three vectors **A**, **B** and **C** which have coincident initial points and are not *coplanar*, i.e. do not lie in or are not parallel to the same plane, are said to form a *right-handed system* or *dextral system* if a right threaded screw rotated through an angle less than 180° from **A** to **B** will advance in the direction **C** as shown in Fig.6.

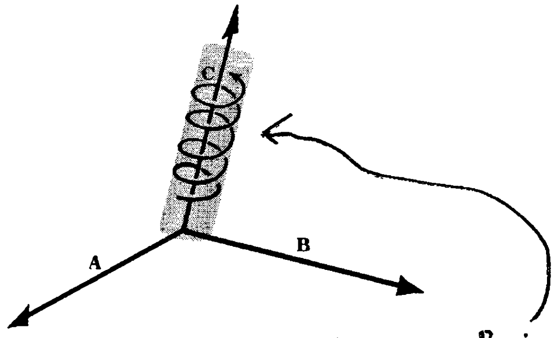



Fig. 6

from A to B

**COMPONENTS OF A VECTOR.** Any vector **A** in 3 dimensions can be represented with initial point at the origin $O$ of a rectangular coordinate system (Fig.7). Let $(A_1, A_2, A_3)$ be the rectangular coordinates of the terminal point of vector **A** with initial point at $O$. The vectors $A_1\mathbf{i}$, $A_2\mathbf{j}$, and $A_3\mathbf{k}$ are called the *rectangular component vectors* or simply *component vectors* of **A** in the $x$, $y$ and $z$ directions respectively. $A_1$, $A_2$ and $A_3$ are called the *rectangular components* or simply *components* of **A** in the $x$, $y$ and $z$ directions respectively.

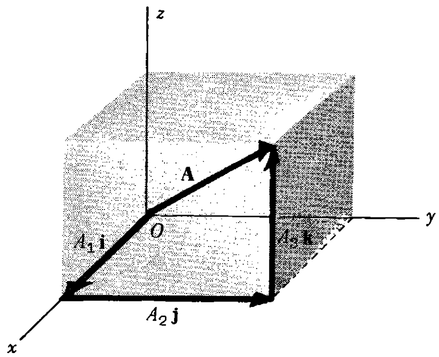



Fig. 7

The sum or resultant of $A_1\mathbf{i}$, $A_2\mathbf{j}$ and $A_3\mathbf{k}$ is the vector **A** so that we can write

$$\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$$

The magnitude of **A** is $$A = |\mathbf{A}| = \sqrt{A_1^2 + A_2^2 + A_3^2}$$

In particular, the *position vector* or *radius vector* **r** from $O$ to the point $(x,y,z)$ is written

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

and has magnitude $r = |\mathbf{r}| = \sqrt{x^2 + y^2 + z^2}$.

**SCALAR FIELD.** If to each point $(x,y,z)$ of a region $R$ in space there corresponds a number or scalar $\phi(x,y,z)$, then $\phi$ is called a *scalar function of position* or *scalar point function* and we say that a *scalar field* $\phi$ has been defined in $R$.

**Examples.** *(1)* The temperature at any point within or on the earth's surface at a certain time defines a scalar field.

*(2)* $\phi(x,y,z) = x^3y - z^2$ defines a scalar field.

A scalar field which is independent of time is called a *stationary* or *steady-state scalar field.*

**VECTOR FIELD.** If to each point $(x,y,z)$ of a region $R$ in space there corresponds a vector $\mathbf{V}(x,y,z)$, then **V** is called a *vector function of position* or *vector point function* and we say that a *vector field* **V** has been defined in $R$.

**Examples.** *(1)* If the velocity at any point $(x,y,z)$ within a moving fluid is known at a certain time, then a vector field is defined.

*(2)* $\mathbf{V}(x,y,z) = xy^2\mathbf{i} - 2yz^3\mathbf{j} + x^2z\,\mathbf{k}$ defines a vector field.

A vector field which is independent of time is called a *stationary* or *steady-state vector field.*

## SOLVED PROBLEMS

1. State which of the following are scalars and which are vectors.

(*a*) weight (*c*) specific heat (*e*) density (*g*) volume (*i*) speed
(*b*) calorie (*d*) momentum (*f*) energy (*h*) distance (*j*) magnetic field intensity

*Ans.* (*a*) vector (*c*) scalar (*e*) scalar (*g*) scalar (*i*) scalar
(*b*) scalar (*d*) vector (*f*) scalar (*h*) scalar (*j*) vector

2. Represent graphically (*a*) a force of 10 lb in a direction 30° north of east
(*b*) a force of 15 lb in a direction 30° east of north.

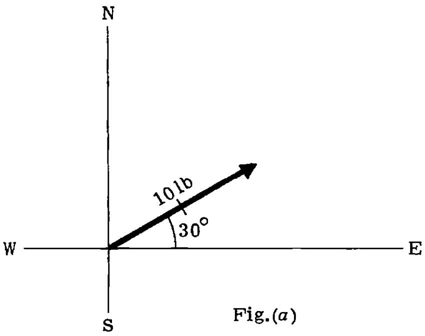



Fig.(*a*)

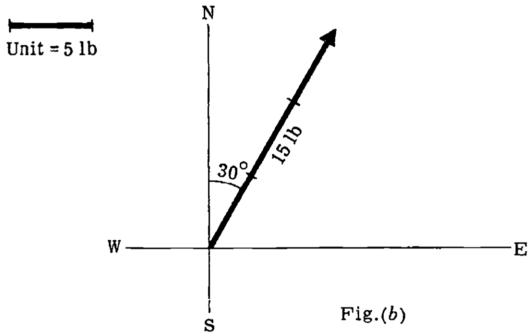



Fig.(*b*)

Choosing the unit of magnitude shown, the required vectors are as indicated above.

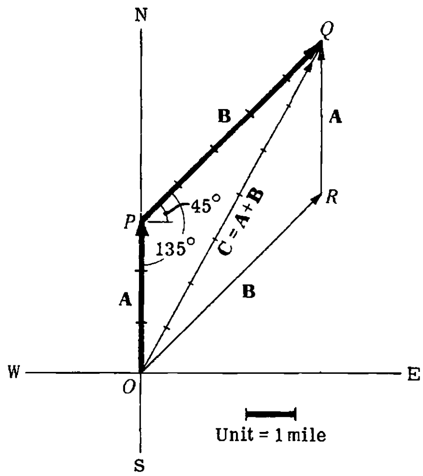
3. An automobile travels 3 miles due north, then 5 miles northeast. Represent these displacements graphically and determine the resultant displacement (*a*) graphically, (*b*) analytically.

Vector **OP** or **A** represents displacement of 3 mi due north.

Vector **PQ** or **B** represents displacement of 5 mi north east.

Vector **OQ** or **C** represents the resultant displacement or sum of vectors **A** and **B**, i.e. **C** = **A**+**B**. This is the *triangle law* of vector addition.

The resultant vector **OQ** can also be obtained by constructing the diagonal of the parallelogram *OPQR* having vectors **OP** = **A** and **OR** (equal to vector **PQ** or **B**) as sides. This is the *parallelogram law* of vector addition.




(*a*) *Graphical Determination of Resultant.* Lay off the 1 mile unit on vector **OQ** to find the magnitude 7.4 mi (approximately). Angle $EOQ = 61.5°$, using a protractor. Then vector **OQ** has magnitude 7.4 mi and direction 61.5° north of east.

(*b*) *Analytical Determination of Resultant.* From triangle *OPQ*, denoting the magnitudes of **A**, **B**, **C** by A, B, C, we have by the law of cosines

$$C^2 = A^2 + B^2 - 2AB\cos\angle OPQ = 3^2 + 5^2 - 2(3)(5)\cos 135° = 34 + 15\sqrt{2} = 55.21$$

and C = 7.43 (approximately).

By the law of sines, $\dfrac{A}{\sin\angle OQP} = \dfrac{C}{\sin\angle OPQ}$. Then

$$\sin \angle OQP = \frac{A \sin \angle OPQ}{C} = \frac{3(0.707)}{7.43} = 0.2855 \quad \text{and} \quad \angle OQP = 16^\circ 35'.$$

Thus vector **OQ** has magnitude 7.43 mi and direction $(45^\circ + 16^\circ 35') = 61^\circ 35'$ north of east.

4. Find the sum or resultant of the following displacements:
**A**, 10 ft northwest; **B**, 20 ft $30^\circ$ north of east; **C**, 35 ft due south. See Fig. (*a*) below.

At the terminal point of **A** place the initial point of **B**.

At the terminal point of **B** place the initial point of **C**.

The resultant **D** is formed by joining the initial point of **A** to the terminal point of **C**, i.e. **D** = **A** + **B** + **C**.

Graphically the resultant is measured to have magnitude of 4.1 units = 20.5 ft and direction $60^\circ$ south of E.

For an analytical method of addition of 3 or more vectors, either in a plane or in space see Problem 26.

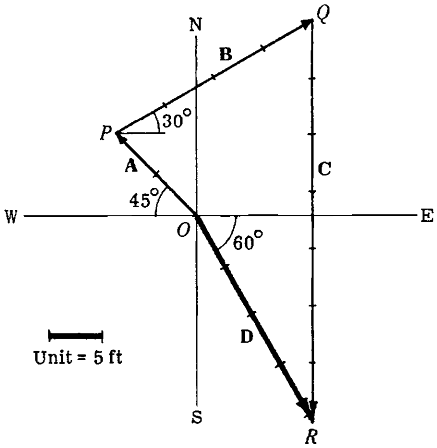



Fig. (*a*)

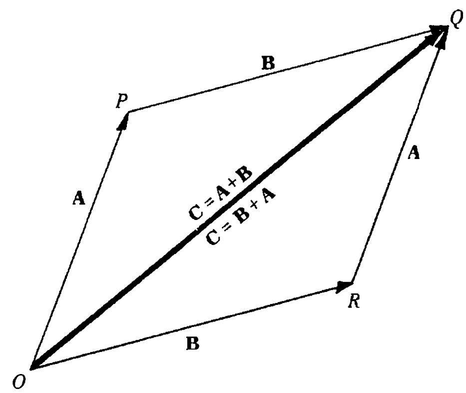



Fig. (*b*)

5. Show that addition of vectors is commutative, i.e. **A** + **B** = **B** + **A**. See Fig. (*b*) above.

$$\mathbf{OP} + \mathbf{PQ} = \mathbf{OQ} \quad \text{or} \quad \mathbf{A} + \mathbf{B} = \mathbf{C},$$
$$\text{and} \quad \mathbf{OR} + \mathbf{RQ} = \mathbf{OQ} \quad \text{or} \quad \mathbf{B} + \mathbf{A} = \mathbf{C}.$$

Then $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$.

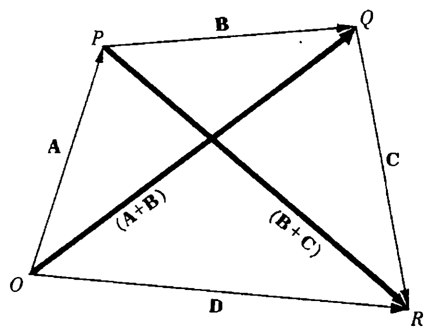
6. Show that the addition of vectors is associative, i.e. **A** + (**B** + **C**) = (**A** + **B**) + **C** .

$$\mathbf{OP} + \mathbf{PQ} = \mathbf{OQ} = (\mathbf{A} + \mathbf{B}),$$
$$\text{and} \quad \mathbf{PQ} + \mathbf{QR} = \mathbf{PR} = (\mathbf{B} + \mathbf{C}).$$

$\mathbf{OP} + \mathbf{PR} = \mathbf{OR} = \mathbf{D}$, i.e. $\mathbf{A} + (\mathbf{B} + \mathbf{C}) = \mathbf{D}$.

$\mathbf{OQ} + \mathbf{QR} = \mathbf{OR} = \mathbf{D}$, i.e. $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{D}$.

Then $\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$.

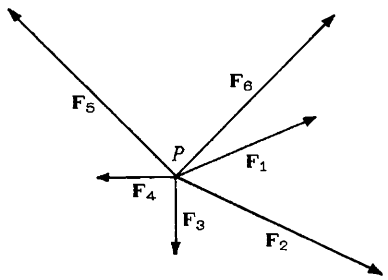


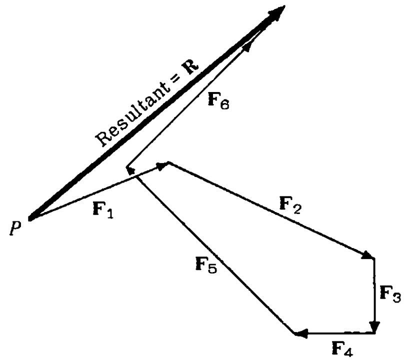

Extensions of the results of Problems 5 and 6 show that the order of addition of any number of vectors is immaterial.

7. Forces $\mathbf{F}_1, \mathbf{F}_2, \ldots, \mathbf{F}_6$ act as shown on object $P$. What force is needed to prevent $P$ from moving ?

Since the order of addition of vectors is immaterial, we may start with any vector, say $\mathbf{F}_1$. To $\mathbf{F}_1$ add $\mathbf{F}_2$, then $\mathbf{F}_3$, etc. The vector drawn from the initial point of $\mathbf{F}_1$ to the terminal point of $\mathbf{F}_6$ is the resultant $\mathbf{R}$, i.e. $\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \mathbf{F}_4 + \mathbf{F}_5 + \mathbf{F}_6$ .

The force needed to prevent $P$ from moving is $-\mathbf{R}$ which is a vector equal in magnitude to $\mathbf{R}$ but opposite in direction and sometimes called the *equilibrant*.




8. Given vectors $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ (Fig.1*a*), construct (*a*) $\mathbf{A} - \mathbf{B} + 2\mathbf{C}$ (*b*) $3\mathbf{C} - \frac{1}{2}(2\mathbf{A} - \mathbf{B})$.

(*a*)

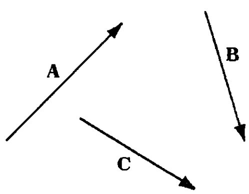



Fig. 1(*a*)

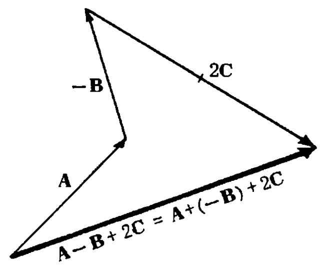



Fig. 2(*a*)

(*b*)

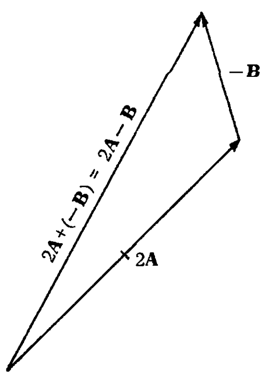



Fig. 1(*b*)

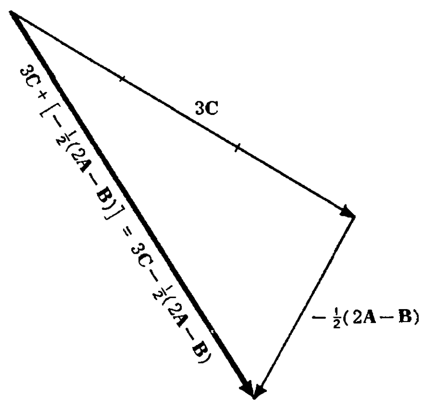
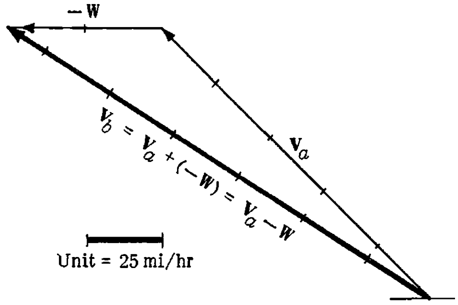



Fig. 2(*b*)

9. An airplane moves in a northwesterly direction at 125 mi/hr relative to the ground, due to the fact there is a westerly wind of 50 mi/hr relative to the ground. How fast and in what direction would the plane have traveled if there were no wind ?




Let $\mathbf{W}$ = wind velocity

$\mathbf{V}_a$ = velocity of plane with wind

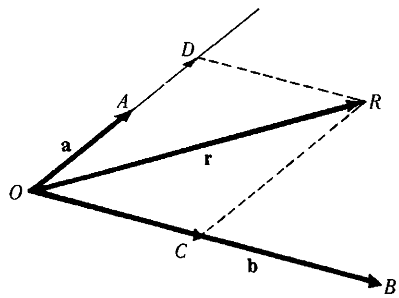
$\mathbf{V}_b$ = velocity of plane without wind

Then $\mathbf{V}_a = \mathbf{V}_b + \mathbf{W}$ or $\mathbf{V}_b = \mathbf{V}_a - \mathbf{W} = \mathbf{V}_a + (-\mathbf{W})$

$\mathbf{V}_b$ has magnitude 6.5 units = 163 mi/hr and direction 33° north of west.

10. Given two non-collinear vectors **a** and **b**, find an expression for any vector **r** lying in the plane determined by **a** and **b**.

Non-collinear vectors are vectors which are not parallel to the same line. Hence when their initial points coincide, they determine a plane. Let **r** be any vector lying in the plane of **a** and **b** and having its initial point coincident with the initial points of **a** and **b** at $O$. From the terminal point $R$ of **r** construct lines parallel to the vectors **a** and **b** and complete the parallelogram $ODRC$ by extension of the lines of action of **a** and **b** if necessary. From the adjoining figure




$$\mathbf{OD} = x(\mathbf{OA}) = x\,\mathbf{a}, \quad \text{where } x \text{ is a scalar}$$
$$\mathbf{OC} = y(\mathbf{OB}) = y\,\mathbf{b}, \quad \text{where } y \text{ is a scalar.}$$

But by the parallelogram law of vector addition

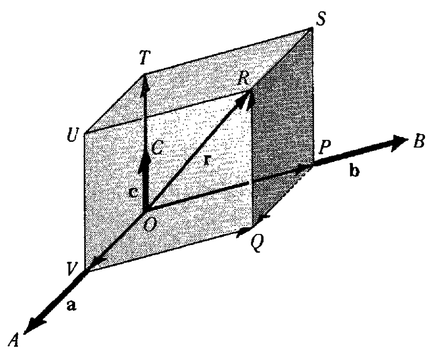
$$\mathbf{OR} = \mathbf{OD} + \mathbf{OC} \quad \text{or} \quad \mathbf{r} = x\,\mathbf{a} + y\,\mathbf{b}$$

which is the required expression. The vectors $x\,\mathbf{a}$ and $y\,\mathbf{b}$ are called *component vectors* of **r** in the directions **a** and **b** respectively. The scalars $x$ and $y$ may be positive or negative depending on the relative orientations of the vectors. From the manner of construction it is clear that $x$ and $y$ are unique for a given **a**, **b**, and **r**. The vectors **a** and **b** are called *base vectors* in a plane.

11. Given three non-coplanar vectors **a**, **b**, and **c**, find an expression for any vector **r** in three dimensional space.

Non-coplanar vectors are vectors which are not parallel to the same plane. Hence when their initial points coincide they do not lie in the same plane.

Let **r** be any vector in space having its initial point coincident with the initial points of **a**, **b** and **c** at $O$. Through the terminal point of **r** pass planes parallel respectively to the planes determined by **a** and **b**, **b** and **c**, and **a** and **c**; and complete the parallelepiped $PQRSTUV$ by extension of the lines of action of **a**, **b** and **c** if necessary. From the adjoining figure,




$$\mathbf{OV} = x(\mathbf{OA}) = x\,\mathbf{a} \quad \text{where } x \text{ is a scalar}$$
$$\mathbf{OP} = y(\mathbf{OB}) = y\,\mathbf{b} \quad \text{where } y \text{ is a scalar}$$
$$\mathbf{OT} = z(\mathbf{OC}) = z\,\mathbf{c} \quad \text{where } z \text{ is a scalar.}$$

But $\mathbf{OR} = \mathbf{OV} + \mathbf{VQ} + \mathbf{QR} = \mathbf{OV} + \mathbf{OP} + \mathbf{OT}$ or $\mathbf{r} = x\,\mathbf{a} + y\,\mathbf{b} + z\,\mathbf{c}$.

From the manner of construction it is clear that $x$, $y$ and $z$ are unique for a given **a**, **b**, **c** and **r**.

The vectors $x\mathbf{a}$, $y\mathbf{b}$ and $z\mathbf{c}$ are called *component vectors* of $\mathbf{r}$ in directions $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively. The vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are called *base vectors* in three dimensions.

As a special case, if $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$ are the unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$, which are mutually perpendicular, we see that any vector $\mathbf{r}$ can be expressed uniquely in terms of $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ by the expression $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

Also, if $\mathbf{c} = \mathbf{0}$ then $\mathbf{r}$ must lie in the plane of $\mathbf{a}$ and $\mathbf{b}$ so the result of Problem 10 is obtained.

**12.** Prove that if $\mathbf{a}$ and $\mathbf{b}$ are non-collinear then $x\mathbf{a} + y\mathbf{b} = \mathbf{0}$ implies $x = y = 0$.

Suppose $x \neq 0$. Then $x\mathbf{a} + y\mathbf{b} = \mathbf{0}$ implies $x\mathbf{a} = -y\mathbf{b}$ or $\mathbf{a} = -(y/x)\mathbf{b}$, i.e. $\mathbf{a}$ and $\mathbf{b}$ must be parallel to to the same line (collinear) contrary to hypothesis. Thus $x = 0$; then $y\mathbf{b} = \mathbf{0}$, from which $y = 0$.

**13.** If $x_1\mathbf{a} + y_1\mathbf{b} = x_2\mathbf{a} + y_2\mathbf{b}$, where $\mathbf{a}$ and $\mathbf{b}$ are non-collinear, then $x_1 = x_2$ and $y_1 = y_2$.

$x_1\mathbf{a} + y_1\mathbf{b} = x_2\mathbf{a} + y_2\mathbf{b}$ can be written

$$x_1\mathbf{a} + y_1\mathbf{b} - (x_2\mathbf{a} + y_2\mathbf{b}) = \mathbf{0} \quad \text{or} \quad (x_1 - x_2)\mathbf{a} + (y_1 - y_2)\mathbf{b} = \mathbf{0}.$$

Hence by Problem 12, $x_1 - x_2 = 0$, $y_1 - y_2 = 0$ or $x_1 = x_2$, $y_1 = y_2$.

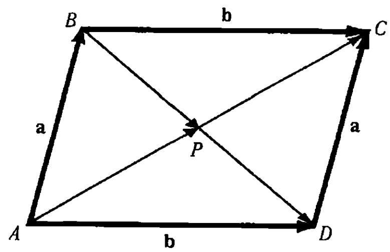
**14.** Prove that if $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are non-coplanar then $x\mathbf{a} + y\mathbf{b} + z\mathbf{c} = \mathbf{0}$ implies $x = y = z = 0$.

Suppose $x \neq 0$. Then $x\mathbf{a} + y\mathbf{b} + z\mathbf{c} = \mathbf{0}$ implies $x\mathbf{a} = -y\mathbf{b} - z\mathbf{c}$ or $\mathbf{a} = -(y/x)\mathbf{b} - (z/x)\mathbf{c}$. But $-(y/x)\mathbf{b} - (z/x)\mathbf{c}$ is a vector lying in the plane of $\mathbf{b}$ and $\mathbf{c}$ (Problem 10), i.e. $\mathbf{a}$ lies in the plane of $\mathbf{b}$ and $\mathbf{c}$ which is clearly a contradiction to the hypothesis that $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are non-coplanar. Hence $x = 0$. By similar reasoning, contradictions are obtained upon supposing $y \neq 0$ and $z \neq 0$.

**15.** If $x_1\mathbf{a} + y_1\mathbf{b} + z_1\mathbf{c} = x_2\mathbf{a} + y_2\mathbf{b} + z_2\mathbf{c}$, where $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are non-coplanar, then $x_1 = x_2$, $y_1 = y_2$, $z_1 = z_2$.

The equation can be written $(x_1 - x_2)\mathbf{a} + (y_1 - y_2)\mathbf{b} + (z_1 - z_2)\mathbf{c} = \mathbf{0}$. Then by Problem 14, $x_1 - x_2 = 0$, $y_1 - y_2 = 0$, $z_1 - z_2 = 0$ or $x_1 = x_2$, $y_1 = y_2$, $z_1 = z_2$.

**16.** Prove that the diagonals of a parallelogram bisect each other.




Let $ABCD$ be the given parallelogram with diagonals intersecting at $P$.

Since $\mathbf{BD} + \mathbf{a} = \mathbf{b}$, $\mathbf{BD} = \mathbf{b} - \mathbf{a}$. Then $\mathbf{BP} = x(\mathbf{b} - \mathbf{a})$.

Since $\mathbf{AC} = \mathbf{a} + \mathbf{b}$, $\mathbf{AP} = y(\mathbf{a} + \mathbf{b})$.

But $\mathbf{AB} = \mathbf{AP} + \mathbf{PB} = \mathbf{AP} - \mathbf{BP}$,
i.e. $\mathbf{a} = y(\mathbf{a} + \mathbf{b}) - x(\mathbf{b} - \mathbf{a}) = (x + y)\mathbf{a} + (y - x)\mathbf{b}$.

Since $\mathbf{a}$ and $\mathbf{b}$ are non-collinear we have by Problem 13, $x + y = 1$ and $y - x = 0$, i.e. $x = y = \frac{1}{2}$ and $P$ is the midpoint of both diagonals.

**17.** If the midpoints of the consecutive sides of any quadrilateral are connected by straight lines, prove that the resulting quadrilateral is a parallelogram.

Let $ABCD$ be the given quadrilateral and $P, Q, R, S$ the midpoints of its sides. Refer to Fig.(*a*) below.

Then $\mathbf{PQ} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$, $\mathbf{QR} = \frac{1}{2}(\mathbf{b} + \mathbf{c})$, $\mathbf{RS} = \frac{1}{2}(\mathbf{c} + \mathbf{d})$, $\mathbf{SP} = \frac{1}{2}(\mathbf{d} + \mathbf{a})$.

But $\mathbf{a} + \mathbf{b} + \mathbf{c} + \mathbf{d} = \mathbf{0}$. Then

$$\mathbf{PQ} = \tfrac{1}{2}(\mathbf{a} + \mathbf{b}) = -\tfrac{1}{2}(\mathbf{c} + \mathbf{d}) = \mathbf{SR} \quad \text{and} \quad \mathbf{QR} = \tfrac{1}{2}(\mathbf{b} + \mathbf{c}) = -\tfrac{1}{2}(\mathbf{d} + \mathbf{a}) = \mathbf{PS}$$

Thus opposite sides are equal and parallel and $PQRS$ is a parallelogram.

**18.** Let $P_1$, $P_2$, $P_3$ be points fixed relative to an origin $O$ and let $\mathbf{r}_1$, $\mathbf{r}_2$, $\mathbf{r}_3$ be position vectors from $O$ to each point. Show that if the vector equation $a_1\mathbf{r}_1 + a_2\mathbf{r}_2 + a_3\mathbf{r}_3 = \mathbf{0}$ holds with respect to origin $O$ then it will hold with respect to any other origin $O'$ if and only if $a_1 + a_2 + a_3 = 0$.

Let $\mathbf{r}_1'$, $\mathbf{r}_2'$ and $\mathbf{r}_3'$ be the position vectors of $P_1$, $P_2$ and $P_3$ with respect to $O'$ and let $\mathbf{v}$ be the position vector of $O'$ with respect to $O$. We seek conditions under which the equation $a_1\mathbf{r}_1' + a_2\mathbf{r}_2' + a_3\mathbf{r}_3' = \mathbf{0}$ will hold in the new reference system.

From Fig.($b$) below, it is clear that $\mathbf{r}_1 = \mathbf{v} + \mathbf{r}_1'$, $\mathbf{r}_2 = \mathbf{v} + \mathbf{r}_2'$, $\mathbf{r}_3 = \mathbf{v} + \mathbf{r}_3'$ so that $a_1\mathbf{r}_1 + a_2\mathbf{r}_2 + a_3\mathbf{r}_3 = \mathbf{0}$ becomes

$$\begin{aligned} a_1\mathbf{r}_1 + a_2\mathbf{r}_2 + a_3\mathbf{r}_3 &= a_1(\mathbf{v} + \mathbf{r}_1') + a_2(\mathbf{v} + \mathbf{r}_2') + a_3(\mathbf{v} + \mathbf{r}_3') \\ &= (a_1 + a_2 + a_3)\mathbf{v} + a_1\mathbf{r}_1' + a_2\mathbf{r}_2' + a_3\mathbf{r}_3' = \mathbf{0} \end{aligned}$$

The result $a_1\mathbf{r}_1' + a_2\mathbf{r}_2' + a_3\mathbf{r}_3' = \mathbf{0}$ will hold if and only if

$$(a_1 + a_2 + a_3)\mathbf{v} = \mathbf{0}, \quad \text{i.e.} \quad a_1 + a_2 + a_3 = 0.$$

The result can be generalized.

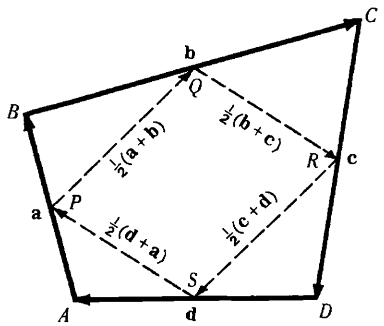



Fig.($a$)

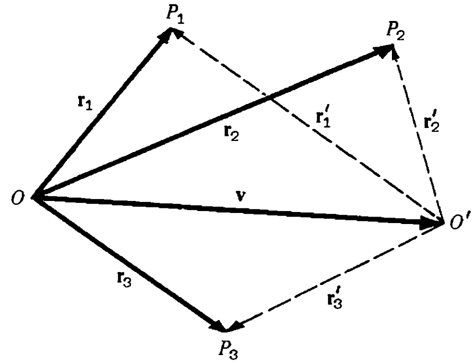



Fig.($b$)

**19.** Find the equation of a straight line which passes through two given points $A$ and $B$ having position vectors $\mathbf{a}$ and $\mathbf{b}$ with respect to an origin $O$.

Let $\mathbf{r}$ be the position vector of any point $P$ on the line through $A$ and $B$.

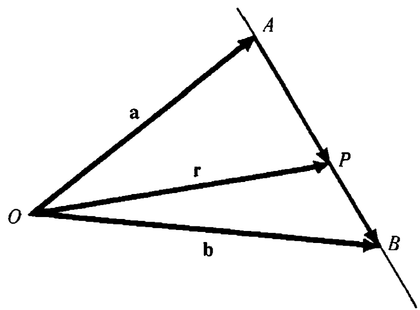
From the adjoining figure,

$$\mathbf{OA} + \mathbf{AP} = \mathbf{OP} \quad \text{or} \quad \mathbf{a} + \mathbf{AP} = \mathbf{r}, \quad \text{i.e.} \quad \mathbf{AP} = \mathbf{r} - \mathbf{a}$$

$$\text{and} \quad \mathbf{OA} + \mathbf{AB} = \mathbf{OB} \quad \text{or} \quad \mathbf{a} + \mathbf{AB} = \mathbf{b}, \quad \text{i.e.} \quad \mathbf{AB} = \mathbf{b} - \mathbf{a}$$

Since $\mathbf{AP}$ and $\mathbf{AB}$ are collinear, $\mathbf{AP} = t\mathbf{AB}$ or $\mathbf{r} - \mathbf{a} = t(\mathbf{b} - \mathbf{a})$. Then the required equation is

$$\mathbf{r} = \mathbf{a} + t(\mathbf{b} - \mathbf{a}) \quad \text{or} \quad \mathbf{r} = (1-t)\mathbf{a} + t\mathbf{b}$$



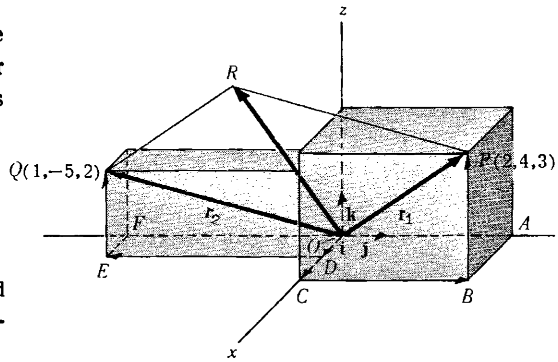

If the equation is written $(1-t)\mathbf{a} + t\mathbf{b} - \mathbf{r} = \mathbf{0}$, the sum of the coefficients of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{r}$ is $1 - t + t - 1 = 0$. Hence by Problem 18 it is seen that the point $P$ is always on the line joining $A$ and $B$ and does not depend on the choice of origin $O$, which is of course as it should be.

*Another Method.* Since $\mathbf{AP}$ and $\mathbf{PB}$ are collinear, we have for scalars $m$ and $n$:

$$m\mathbf{AP} = n\mathbf{PB} \quad \text{or} \quad m(\mathbf{r} - \mathbf{a}) = n(\mathbf{b} - \mathbf{r})$$

Solving, $\mathbf{r} = \dfrac{m\mathbf{a} + n\mathbf{b}}{m + n}$ which is called the *symmetric form*.

**20.** (*a*) Find the position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ for the points $P(2,4,3)$ and $Q(1,-5,2)$ of a rectangular coordinate system in terms of the unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$. (*b*) Determine graphically and analytically the resultant of these position vectors.



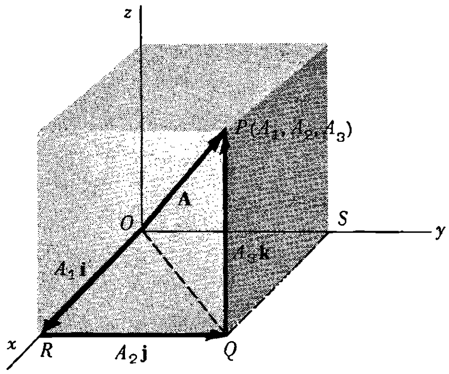

(*a*) $\mathbf{r}_1 = \mathbf{OP} = \mathbf{OC} + \mathbf{CB} + \mathbf{BP} = 2\mathbf{i} + 4\mathbf{j} + 3\mathbf{k}$

$\mathbf{r}_2 = \mathbf{OQ} = \mathbf{OD} + \mathbf{DE} + \mathbf{EQ} = \mathbf{i} - 5\mathbf{j} + 2\mathbf{k}$

(*b*) *Graphically*, the resultant of $\mathbf{r}_1$ and $\mathbf{r}_2$ is obtained as the diagonal **OR** of parallelogram $OPRQ$. *Analytically*, the resultant of $\mathbf{r}_1$ and $\mathbf{r}_2$ is given by

$$\mathbf{r}_1 + \mathbf{r}_2 = (2\mathbf{i}+4\mathbf{j}+3\mathbf{k}) + (\mathbf{i}-5\mathbf{j}+2\mathbf{k}) = 3\mathbf{i} - \mathbf{j} + 5\mathbf{k}$$

**21.** Prove that the magnitude $A$ of the vector $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$ is $A = \sqrt{A_1^2 + A_2^2 + A_3^2}$.




By the Pythagorean theorem,

$$(\overline{OP})^2 = (\overline{OQ})^2 + (\overline{QP})^2$$

where $\overline{OP}$ denotes the magnitude of vector **OP**, etc. Similarly, $(\overline{OQ})^2 = (\overline{OR})^2 + (\overline{RQ})^2$.

Then $(\overline{OP})^2 = (\overline{OR})^2 + (\overline{RQ})^2 + (\overline{QP})^2$ or

$A^2 = A_1^2 + A_2^2 + A_3^2$, i.e. $A = \sqrt{A_1^2 + A_2^2 + A_3^2}$.

**22.** Given $\mathbf{r}_1 = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{r}_2 = 2\mathbf{i} - 4\mathbf{j} - 3\mathbf{k}$, $\mathbf{r}_3 = -\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$, find the magnitudes of (*a*) $\mathbf{r}_3$, (*b*) $\mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3$, (*c*) $2\mathbf{r}_1 - 3\mathbf{r}_2 - 5\mathbf{r}_3$.

(*a*) $|\mathbf{r}_3| = |-\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}| = \sqrt{(-1)^2 + (2)^2 + (2)^2} = 3$.

(*b*) $\mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3 = (3\mathbf{i} - 2\mathbf{j} + \mathbf{k}) + (2\mathbf{i} - 4\mathbf{j} - 3\mathbf{k}) + (-\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}) = 4\mathbf{i} - 4\mathbf{j} + 0\mathbf{k} = 4\mathbf{i} - 4\mathbf{j}$

Then $|\mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3| = |4\mathbf{i} - 4\mathbf{j} + 0\mathbf{k}| = \sqrt{(4)^2 + (-4)^2 + (0)^2} = \sqrt{32} = 4\sqrt{2}$.

(*c*) $2\mathbf{r}_1 - 3\mathbf{r}_2 - 5\mathbf{r}_3 = 2(3\mathbf{i} - 2\mathbf{j} + \mathbf{k}) - 3(2\mathbf{i} - 4\mathbf{j} - 3\mathbf{k}) - 5(-\mathbf{i} + 2\mathbf{j} + 2\mathbf{k})$

$= 6\mathbf{i} - 4\mathbf{j} + 2\mathbf{k} - 6\mathbf{i} + 12\mathbf{j} + 9\mathbf{k} + 5\mathbf{i} - 10\mathbf{j} - 10\mathbf{k} = 5\mathbf{i} - 2\mathbf{j} + \mathbf{k}$.

Then $|2\mathbf{r}_1 - 3\mathbf{r}_2 - 5\mathbf{r}_3| = |5\mathbf{i} - 2\mathbf{j} + \mathbf{k}| = \sqrt{(5)^2 + (-2)^2 + (1)^2} = \sqrt{30}$.

**23.** If $\mathbf{r}_1 = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{r}_2 = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$, $\mathbf{r}_3 = -2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ and $\mathbf{r}_4 = 3\mathbf{i} + 2\mathbf{j} + 5\mathbf{k}$, find scalars $a,b,c$ such that $\mathbf{r}_4 = a\mathbf{r}_1 + b\mathbf{r}_2 + c\mathbf{r}_3$.

We require $3\mathbf{i} + 2\mathbf{j} + 5\mathbf{k} = a(2\mathbf{i} - \mathbf{j} + \mathbf{k}) + b(\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}) + c(-2\mathbf{i} + \mathbf{j} - 3\mathbf{k})$

$= (2a + b - 2c)\mathbf{i} + (-a + 3b + c)\mathbf{j} + (a - 2b - 3c)\mathbf{k}$.

Since $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are non-coplanar we have by Problem 15,

$$2a + b - 2c = 3, \quad -a + 3b + c = 2, \quad a - 2b - 3c = 5.$$

Solving, $a = -2$, $b = 1$, $c = -3$ and $\mathbf{r}_4 = -2\mathbf{r}_1 + \mathbf{r}_2 - 3\mathbf{r}_3$.

The vector $\mathbf{r}_4$ is said to be *linearly dependent* on $\mathbf{r}_1$, $\mathbf{r}_2$, and $\mathbf{r}_3$; in other words $\mathbf{r}_1$, $\mathbf{r}_2$, $\mathbf{r}_3$ and $\mathbf{r}_4$ constitute a *linearly dependent* set of vectors. On the other hand any three (or fewer) of these vectors are *linearly independent*.

In general the vectors $\mathbf{A}, \mathbf{B}, \mathbf{C}, \ldots$ are called linearly dependent if we can find a set of scalars, $a, b, c, \ldots$, not all zero, so that $a\mathbf{A} + b\mathbf{B} + c\mathbf{C} + \ldots = \mathbf{0}$, otherwise they are linearly independent.

**24.** Find a unit vector parallel to the resultant of vectors $\mathbf{r}_1 = 2\mathbf{i} + 4\mathbf{j} - 5\mathbf{k}$, $\mathbf{r}_2 = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$.

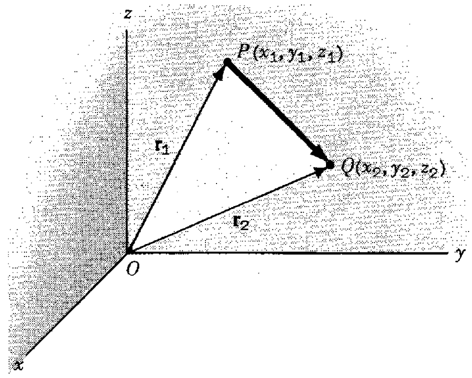
Resultant $\mathbf{R} = \mathbf{r}_1 + \mathbf{r}_2 = (2\mathbf{i}+4\mathbf{j}-5\mathbf{k}) + (\mathbf{i}+2\mathbf{j}+3\mathbf{k}) = 3\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}$.

$R = |\mathbf{R}| = |3\mathbf{i}+6\mathbf{j}-2\mathbf{k}| = \sqrt{(3)^2+(6)^2+(-2)^2} = 7$.

Then a unit vector parallel to $\mathbf{R}$ is $\dfrac{\mathbf{R}}{R} = \dfrac{3\mathbf{i}+6\mathbf{j}-2\mathbf{k}}{7} = \dfrac{3}{7}\mathbf{i} + \dfrac{6}{7}\mathbf{j} - \dfrac{2}{7}\mathbf{k}$.

Check: $\left|\frac{3}{7}\mathbf{i} + \frac{6}{7}\mathbf{j} - \frac{2}{7}\mathbf{k}\right| = \sqrt{(\frac{3}{7})^2 + (\frac{6}{7})^2 + (-\frac{2}{7})^2} = 1$.

**25.** Determine the vector having initial point $P(x_1, y_1, z_1)$ and terminal point $Q(x_2, y_2, z_2)$ and find its magnitude.




The position vector of $P$ is $\mathbf{r}_1 = x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k}$.

The position vector of $Q$ is $\mathbf{r}_2 = x_2\mathbf{i} + y_2\mathbf{j} + z_2\mathbf{k}$.

$\mathbf{r}_1 + \mathbf{PQ} = \mathbf{r}_2$ or

$$\mathbf{PQ} = \mathbf{r}_2 - \mathbf{r}_1 = (x_2\mathbf{i}+y_2\mathbf{j}+z_2\mathbf{k}) - (x_1\mathbf{i}+y_1\mathbf{j}+z_1\mathbf{k})$$
$$= (x_2-x_1)\mathbf{i} + (y_2-y_1)\mathbf{j} + (z_2-z_1)\mathbf{k}.$$

Magnitude of $\mathbf{PQ} = \overline{PQ} = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2}$.

Note that this is the distance between points $P$ and $Q$.

**26.** Forces $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ acting on an object are given in terms of their components by the vector equations $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$, $\mathbf{C} = C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k}$. Find the magnitude of the resultant of these forces.

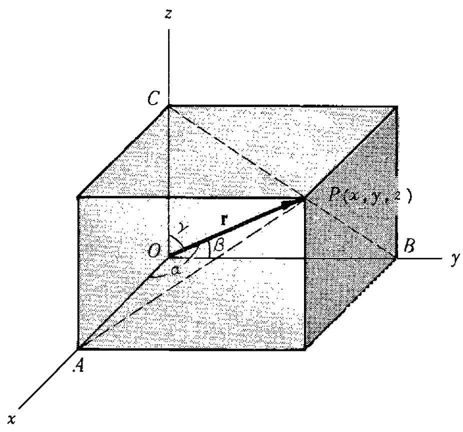
Resultant force $\mathbf{R} = \mathbf{A}+\mathbf{B}+\mathbf{C} = (A_1+B_1+C_1)\mathbf{i} + (A_2+B_2+C_2)\mathbf{j} + (A_3+B_3+C_3)\mathbf{k}$.

Magnitude of resultant $= \sqrt{(A_1+B_1+C_1)^2 + (A_2+B_2+C_2)^2 + (A_3+B_3+C_3)^2}$.

The result is easily extended to more than three forces.

**27.** Determine the angles $\alpha, \beta$ and $\gamma$ which the vector $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ makes with the positive directions of the coordinate axes and show that

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1.$$




Referring to the figure, triangle $OAP$ is a right triangle with right angle at $A$; then $\cos\alpha = \dfrac{x}{|\mathbf{r}|}$. Similarly from right triangles $OBP$ and $OCP$, $\cos\beta = \dfrac{y}{|\mathbf{r}|}$ and $\cos\gamma = \dfrac{z}{|\mathbf{r}|}$. Also, $|\mathbf{r}| = r = \sqrt{x^2+y^2+z^2}$.

Then $\cos\alpha = \dfrac{x}{r}$, $\cos\beta = \dfrac{y}{r}$, $\cos\gamma = \dfrac{z}{r}$ from which $\alpha, \beta, \gamma$ can be obtained. From these it follows that

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{x^2+y^2+z^2}{r^2} = 1.$$

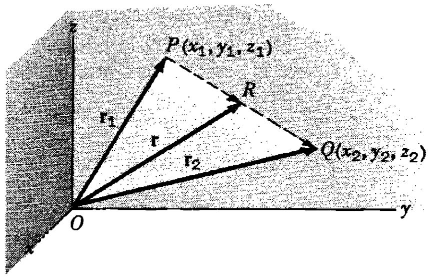
The numbers $\cos\alpha$, $\cos\beta$, $\cos\gamma$ are called the *direction cosines* of the vector $\mathbf{OP}$.

**28.** Determine a set of equations for the straight line passing through the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$.

Let $\mathbf{r}_1$ and $\mathbf{r}_2$ be the position vectors of $P$ and $Q$ respectively, and $\mathbf{r}$ the position vector of any point $R$ on the line joining $P$ and $Q$.

$$\mathbf{r}_1 + \mathbf{PR} = \mathbf{r} \quad \text{or} \quad \mathbf{PR} = \mathbf{r} - \mathbf{r}_1$$
$$\mathbf{r}_1 + \mathbf{PQ} = \mathbf{r}_2 \quad \text{or} \quad \mathbf{PQ} = \mathbf{r}_2 - \mathbf{r}_1$$

But $\mathbf{PR} = t\,\mathbf{PQ}$ where $t$ is a scalar. Then $\mathbf{r} - \mathbf{r}_1 = t(\mathbf{r}_2 - \mathbf{r}_1)$ is the required vector equation of the straight line (compare with Problem 19).




In rectangular coordinates we have, since $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$,

$$(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) - (x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k}) = t\left[(x_2\mathbf{i} + y_2\mathbf{j} + z_2\mathbf{k}) - (x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k})\right]$$

or

$$(x - x_1)\mathbf{i} + (y - y_1)\mathbf{j} + (z - z_1)\mathbf{k} = t\left[(x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}\right]$$

Since $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are non-coplanar vectors we have by Problem 15,

$$x - x_1 = t(x_2 - x_1), \quad y - y_1 = t(y_2 - y_1), \quad z - z_1 = t(z_2 - z_1)$$

as the parametric equations of the line, $t$ being the parameter. Eliminating $t$, the equations become

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}.$$

**29.** Given the scalar field defined by $\phi(x,y,z) = 3x^2z - xy^3 + 5$, find $\phi$ at the points (*a*) $(0,0,0)$, (*b*) $(1,-2,2)$ (*c*) $(-1,-2,-3)$.

(*a*) $\phi(0,0,0) = 3(0)^2(0) - (0)(0)^3 + 5 = 0 - 0 + 5 = 5$

(*b*) $\phi(1,-2,2) = 3(1)^2(2) - (1)(-2)^3 + 5 = 6 + 8 + 5 = 19$

(*c*) $\phi(-1,-2,-3) = 3(-1)^2(-3) - (-1)(-2)^3 + 5 = -9 - 8 + 5 = -12$

**30.** Graph the vector fields defined by:
(*a*) $\mathbf{V}(x,y) = x\mathbf{i} + y\mathbf{j}$, (*b*) $\mathbf{V}(x,y) = -x\mathbf{i} - y\mathbf{j}$, (*c*) $\mathbf{V}(x,y,z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

(*a*) At each point $(x,y)$, except $(0,0)$, of the $xy$ plane there is defined a unique vector $x\mathbf{i} + y\mathbf{j}$ of magnitude $\sqrt{x^2 + y^2}$ having direction passing through the origin and outward from it. To simplify graphing procedures, note that all vectors associated with points on the circles $x^2 + y^2 = a^2$, $a > 0$ have magnitude $a$. The field therefore appears as in Figure ($a$) where an appropriate scale is used.

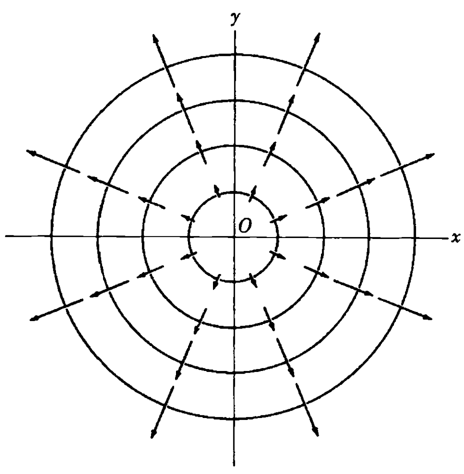



Fig.(*a*)

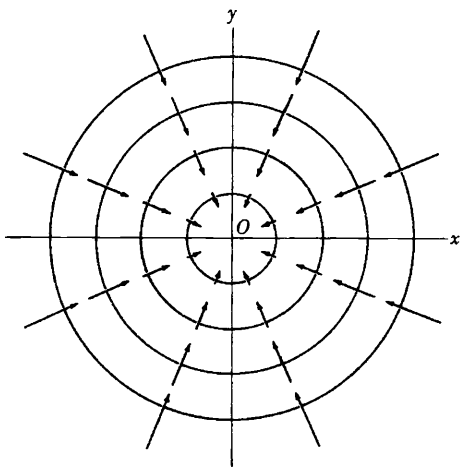



Fig.(*b*)

(*b*) Here each vector is equal to but opposite in direction to the corresponding one in (*a*). The field therefore appears as in Fig.(*b*).

In Fig.(*a*) the field has the appearance of a fluid emerging from a point source $O$ and flowing in the directions indicated. For this reason the field is called a *source field* and $O$ is a *source*.

In Fig.(*b*) the field seems to be flowing toward $O$, and the field is therefore called a *sink field* and $O$ is a *sink*.

In three dimensions the corresponding interpretation is that a fluid is emerging radially from (or proceeding radially toward) a line source (or line sink).

The vector field is called two dimensional since it is independent of $z$.

(*c*) Since the magnitude of each vector is $\sqrt{x^2+y^2+z^2}$, all points on the sphere $x^2+y^2+z^2 = a^2$, $a > 0$ have vectors of magnitude $a$ associated with them. The field therefore takes on the appearance of that of a fluid emerging from source $O$ and proceeding in all directions in space. This is a *three dimensional source field*.

## SUPPLEMENTARY PROBLEMS

**31.** Which of the following are scalars and which are vectors? (*a*) Kinetic energy, (*b*) electric field intensity, (*c*) entropy, (*d*) work, (*e*) centrifugal force, (*f*) temperature, (*g*) gravitational potential, (*h*) charge, (*i*) shearing stress, (*j*) frequency.
*Ans.* (*a*) scalar, (*b*) vector, (*c*) scalar, (*d*) scalar, (*e*) vector, (*f*) scalar, (*g*) scalar, (*h*) scalar, (*i*) vector (*j*) scalar

**32.** An airplane travels 200 miles due west and then 150 miles $60^\circ$ north of west. Determine the resultant displacement (*a*) graphically, (*b*) analytically.
*Ans.* magnitude 304.1 mi ($50\sqrt{37}$), direction $25^\circ 17'$ north of east (arc sin $3\sqrt{111}/74$)

**33.** Find the resultant of the following displacements: **A**, 20 miles $30^\circ$ south of east; **B**, 50 miles due west; **C**, 40 miles northeast; **D**, 30 miles $60^\circ$ south of west.
*Ans.* magnitude 20.9 mi, direction $21^\circ 39'$ south of west

**34.** Show graphically that $-(\mathbf{A}-\mathbf{B}) = -\mathbf{A}+\mathbf{B}$.

**35.** An object $P$ is acted upon by three coplanar forces as shown in Fig.(*a*) below. Determine the force needed to prevent $P$ from moving. *Ans.* 323 lb directly opposite 150 lb force

**36.** Given vectors **A**, **B**, **C** and **D** (Fig.(*b*) below). Construct (*a*) $3\mathbf{A}-2\mathbf{B}-(\mathbf{C}-\mathbf{D})$ (*b*) $\frac{1}{2}\mathbf{C}+\frac{2}{3}(\mathbf{A}-\mathbf{B}+2\mathbf{D})$.

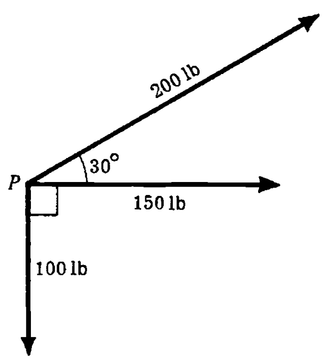



Fig.(*a*)

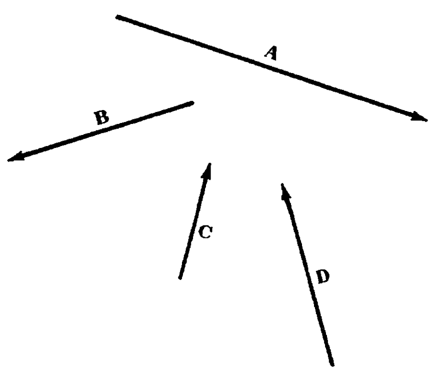



Fig.(*b*)

**37.** If $ABCDEF$ are the vertices of a regular hexagon, find the resultant of the forces represented by the vectors $\mathbf{AB}, \mathbf{AC}, \mathbf{AD}, \mathbf{AE}$ and $\mathbf{AF}$. *Ans.* $3\,\mathbf{AD}$

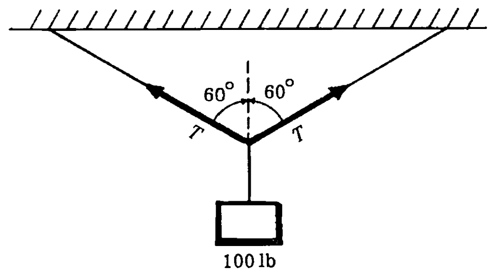
**38.** If $\mathbf{A}$ and $\mathbf{B}$ are given vectors show that (*a*) $|\mathbf{A}+\mathbf{B}| \leqq |\mathbf{A}| + |\mathbf{B}|$, (*b*) $|\mathbf{A}-\mathbf{B}| \geqq |\mathbf{A}| - |\mathbf{B}|$.

**39.** Show that $|\mathbf{A}+\mathbf{B}+\mathbf{C}| \leqq |\mathbf{A}| + |\mathbf{B}| + |\mathbf{C}|$.

**40.** Two towns $A$ and $B$ are situated directly opposite each other on the banks of a river whose width is 8 miles and which flows at a speed of 4 mi/hr. A man located at $A$ wishes to reach town $C$ which is 6 miles upstream from and on the same side of the river as town $B$. If his boat can travel at a maximum speed of 10 mi/hr and if he wishes to reach $C$ in the shortest possible time what course must he follow and how long will the trip take?
*Ans.* A straight line course upstream making an angle of $34^\circ 28'$ with the shore line. 1 hr 25 min.

**41.** A man travelling southward at 15 mi/hr observes that the wind appears to be coming from the west. On increasing his speed to 25 mi/hr it appears to be coming from the southwest. Find the direction and speed of the wind. *Ans.* The wind is coming from a direction $56^\circ 18'$ north of west at 18 mi/hr.

**42.** A 100 lb weight is suspended from the center of a rope as shown in the adjoining figure. Determine the tension $T$ in the rope. *Ans.* 100 lb




**43.** Simplify $2\mathbf{A}+\mathbf{B}+3\mathbf{C} - \{\mathbf{A}-2\mathbf{B}-2(2\mathbf{A}-3\mathbf{B}-\mathbf{C})\}$.
*Ans.* $5\mathbf{A}-3\mathbf{B}+\mathbf{C}$

**44.** If $\mathbf{a}$ and $\mathbf{b}$ are non-collinear vectors and $\mathbf{A} = (x+4y)\mathbf{a} + (2x+y+1)\mathbf{b}$ and $\mathbf{B} = (y-2x+2)\mathbf{a} + (2x-3y-1)\mathbf{b}$, find $x$ and $y$ such that $3\mathbf{A} = 2\mathbf{B}$.
*Ans.* $x = 2$, $y = -1$

**45.** The base vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ are given in terms of the base vectors $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ by the relations

$$\mathbf{a}_1 = 2\mathbf{b}_1 + 3\mathbf{b}_2 - \mathbf{b}_3, \quad \mathbf{a}_2 = \mathbf{b}_1 - 2\mathbf{b}_2 + 2\mathbf{b}_3, \quad \mathbf{a}_3 = -2\mathbf{b}_1 + \mathbf{b}_2 - 2\mathbf{b}_3$$

If $\mathbf{F} = 3\mathbf{b}_1 - \mathbf{b}_2 + 2\mathbf{b}_3$, express $\mathbf{F}$ in terms of $\mathbf{a}_1, \mathbf{a}_2$ and $\mathbf{a}_3$. *Ans.* $2\mathbf{a}_1 + 5\mathbf{a}_2 + 3\mathbf{a}_3$

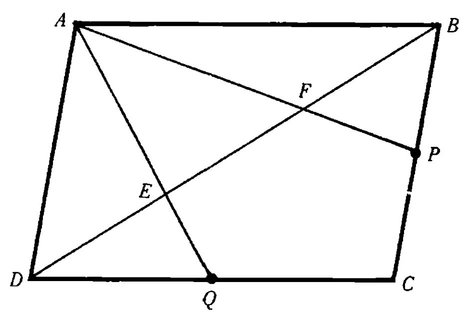
**46.** If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are non-coplanar vectors determine whether the vectors $\mathbf{r}_1 = 2\mathbf{a}-3\mathbf{b}+\mathbf{c}$, $\mathbf{r}_2 = 3\mathbf{a}-5\mathbf{b}+2\mathbf{c}$, and $\mathbf{r}_3 = 4\mathbf{a}-5\mathbf{b}+\mathbf{c}$ are linearly independent or dependent. *Ans.* Linearly dependent since $\mathbf{r}_3 = 5\mathbf{r}_1 - 2\mathbf{r}_2$.

**47.** If $\mathbf{A}$ and $\mathbf{B}$ are given vectors representing the diagonals of a parallelogram, construct the parallelogram.

**48.** Prove that the line joining the midpoints of two sides of a triangle is parallel to the third side and has one half of its magnitude.

**49.** (*a*) If $O$ is any point within triangle $ABC$ and $P, Q, R$ are midpoints of the sides $AB, BC, CA$ respectively, prove that $\mathbf{OA}+\mathbf{OB}+\mathbf{OC} = \mathbf{OP}+\mathbf{OQ}+\mathbf{OR}$.
(*b*) Does the result hold if $O$ is any point outside the triangle? Prove your result. *Ans.* Yes

**50.** In the adjoining figure, $ABCD$ is a parallelogram with $P$ and $Q$ the midpoints of sides $BC$ and $CD$ respectively. Prove that $AP$ and $AQ$ trisect diagonal $BD$ at the points $E$ and $F$.




**51.** Prove that the medians of a triangle meet in a common point which is a point of trisection of the medians.

**52.** Prove that the angle bisectors of a triangle meet in a common point.

**53.** Show that there exists a triangle with sides which are equal and parallel to the medians of any given triangle.

**54.** Let the position vectors of points $P$ and $Q$ relative to an origin $O$ be given by $\mathbf{p}$ and $\mathbf{q}$ respectively. If $R$ is a point which divides line $PQ$ into segments which are in the ratio $m:n$ show that the position vector of $R$

is given by $\mathbf{r} = \dfrac{m\mathbf{p} + n\mathbf{q}}{m+n}$ and that this is independent of the origin.

55. If $\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_n$ are the position vectors of masses $m_1, m_2, \ldots, m_n$ respectively relative to an origin $O$, show that the position vector of the centroid is given by

$$\mathbf{r} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2 + \ldots + m_n\mathbf{r}_n}{m_1 + m_2 + \ldots + m_n}$$

and that this is independent of the origin.

56. A quadrilateral $ABCD$ has masses of 1, 2, 3 and 4 units located respectively at its vertices $A(-1,-2,2)$, $B(3,2,-1)$, $C(1,-2,4)$, and $D(3,1,2)$. Find the coordinates of the centroid. *Ans.* $(2,0,2)$

57. Show that the equation of a plane which passes through three given points $A, B, C$ not in the same straight line and having position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ relative to an origin $O$, can be written

$$\mathbf{r} = \frac{m\mathbf{a} + n\mathbf{b} + p\mathbf{c}}{m + n + p}$$

where $m, n, p$ are scalars. Verify that the equation is independent of the origin.

58. The position vectors of points $P$ and $Q$ are given by $\mathbf{r}_1 = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$, $\mathbf{r}_2 = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$. Determine $\mathbf{PQ}$ in terms of $\mathbf{i}, \mathbf{j}, \mathbf{k}$ and find its magnitude. *Ans.* $2\mathbf{i} - 6\mathbf{j} + 3\mathbf{k}$, 7

59. If $\mathbf{A} = 3\mathbf{i} - \mathbf{j} - 4\mathbf{k}$, $\mathbf{B} = -2\mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$, $\mathbf{C} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, find
(*a*) $2\mathbf{A} - \mathbf{B} + 3\mathbf{C}$, (*b*) $|\mathbf{A} + \mathbf{B} + \mathbf{C}|$, (*c*) $|3\mathbf{A} - 2\mathbf{B} + 4\mathbf{C}|$, (*d*) a unit vector parallel to $3\mathbf{A} - 2\mathbf{B} + 4\mathbf{C}$.
*Ans.* (*a*) $11\mathbf{i} - 8\mathbf{k}$ (*b*) $\sqrt{93}$ (*c*) $\sqrt{398}$ (*d*) $\dfrac{3\mathbf{A} - 2\mathbf{B} + 4\mathbf{C}}{\sqrt{398}}$

60. The following forces act on a particle $P$: $\mathbf{F}_1 = 2\mathbf{i} + 3\mathbf{j} - 5\mathbf{k}$, $\mathbf{F}_2 = -5\mathbf{i} + \mathbf{j} + 3\mathbf{k}$, $\mathbf{F}_3 = \mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$, $\mathbf{F}_4 = 4\mathbf{i} - 3\mathbf{j} - 2\mathbf{k}$, measured in pounds. Find (*a*) the resultant of the forces, (*b*) the magnitude of the resultant.
*Ans.* (*a*) $2\mathbf{i} - \mathbf{j}$ (*b*) $\sqrt{5}$

61. In each case determine whether the vectors are linearly independent or linearly dependent:
(*a*) $\mathbf{A} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$, $\mathbf{B} = \mathbf{i} - 4\mathbf{k}$, $\mathbf{C} = 4\mathbf{i} + 3\mathbf{j} - \mathbf{k}$, (*b*) $\mathbf{A} = \mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$, $\mathbf{B} = 2\mathbf{i} - 4\mathbf{j} - \mathbf{k}$, $\mathbf{C} = 3\mathbf{i} + 2\mathbf{j} - \mathbf{k}$.
*Ans.* (*a*) linearly dependent, (*b*) linearly independent

62. Prove that any four vectors in three dimensions must be linearly dependent.

63. Show that a necessary and sufficient condition that the vectors $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$, $\mathbf{C} = C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k}$ be linearly independent is that the determinant $\begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}$ be different from zero.

64. (*a*) Prove that the vectors $\mathbf{A} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$, $\mathbf{B} = -\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$, $\mathbf{C} = 4\mathbf{i} - 2\mathbf{j} - 6\mathbf{k}$ can form the sides of a triangle.
(*b*) Find the lengths of the medians of the triangle.
*Ans.* (*b*) $\sqrt{6}$, $\frac{1}{2}\sqrt{114}$, $\frac{1}{2}\sqrt{150}$

65. Given the scalar field defined by $\phi(x,y,z) = 4yz^3 + 3xyz - z^2 + 2$. Find (*a*) $\phi(1,-1,-2)$, (*b*) $\phi(0,-3,1)$.
*Ans.* (*a*) 36 (*b*) $-11$

66. Graph the vector fields defined by
(*a*) $\mathbf{V}(x,y) = x\mathbf{i} - y\mathbf{j}$, (*b*) $\mathbf{V}(x,y) = y\mathbf{i} - x\mathbf{j}$, (*c*) $\mathbf{V}(x,y,z) = \dfrac{x\mathbf{i} + y\mathbf{i} + z\mathbf{k}}{\sqrt{x^2 + y^2 + z^2}}$.